Algebra Monsoon 2026 Lecture 2

Subs, products

Def Let M be a magma. A subset $S \subset M$ is a submagma if

$$x, y \in S \Longrightarrow xy \in S$$

If so, S can be viewed as a magma.

Ex. $S = \emptyset$ or $S = M$.

Def Let M be a monoid. A subset $S \subset M$ is a submonoid of

1) S is a submagma
2) $e \in S$.

If so, S can be viewed as a monoid.

Ex. $S = \{e\}$ or $S = M$

For every $m \in \mathbb{N}$, the subset $m\mathbb{N}$ is a submonoid of $\mathbb{N}$.

Def Let G be a group. A subset $H \subset G$ is a subgroup if

1) H is a submonoid
2) $x \in H \longrightarrow x^{-1} \in H$

If so, H can be viewed as a group

Ex. $H = \{e\}$ or $H = G$.

For every $n \in \mathbb{Z}$, the subset $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ (under addition)

For every monoid M, the set of invertible elements $M^\times$ is a submonoid which is a group.

- If $f : M \to N$ is a morphism of monoids, then $f_*(M^\times) \subset N^\times$.
- If $M = \mathrm{End}(E)$ for some set E, then the group $G_E = \mathrm{End}(E)^\times$ is called the symmetric group on E

Proposition Let M be a magma (resp. monoid, resp. group) and let $(S_\lambda)_{\lambda \in L}$ be a family of submagmas (resp. submonoids, resp. subgroups). Then the intersection

$$\bigcap_{\lambda \in L} S_\lambda$$

is a submagma (resp. submonoid, resp. subgroup).

Proof Klar.

If $L = \emptyset$, then $\bigcap_{\lambda \in L} S_\lambda = M$

Def Let M be a magma (resp. monoid, resp. group) and $T \subset M$ a subset. The intersection of all submagmas (resp. submonoids, resp. subgroups) of M containing T is called the submagma (resp. submonoid, resp. subgroup) generated by T and denoted $\langle T \rangle$.

If $T = M$, then $\langle T \rangle = M$.

If M is a magma and $T = \emptyset$, then $\langle T \rangle = \emptyset$.

If M is a monoid or group and $T = \emptyset$, then $\langle T \rangle = \{e\}$.

If $M = \mathbb{Z}$ and $T = \{1\}$
then the submonoid generated by T is $\mathbb{N}$ whereas the subgroup generated by T is $\mathbb{Z}$.
(so the notation $\langle T \rangle$ depends on the context)

If $M = M$, then

(re the notation $\langle T \rangle$ / $\mathrm{sg}(T)$)

If M is a monoid and $x \in M$, then
the submonoid generated by $T = \{x\}$
is the image of the unique morphism
of monoids $j : \mathbb{N} \to M$ such that $j(1) = x$.
We may write $\langle T \rangle = x^{\mathbb{N}}$.

If M is a ~~monoid~~ group and $x \in M$, then
the ~~submonoid~~ subgroup $\langle T \rangle$ generated by $T = \{x\}$
is the image of the unique morphism
of ~~monoids~~ group $j : \mathbb{N} \to M$ such that ~~$j(1) = x$~~ $\tilde{j}(1) = x$.
We may write ~~$\langle T \rangle = x^{\mathbb{N}}$~~ $\langle T \rangle = x^{\mathbb{Z}}$.

In short, $x^{\mathbb{Z}}$ is the subgroup of M
generated by x.

Proposition If $f: M \to N$ is a morphism of magmas (resp. monoids, resp. groups), then $f_*(M)$ is a submagma (resp. submonoid, (resp. subgroup) of N.

Proof Let $x, y \in f_*(M)$. There exist $a, b \in M$ such that $f(a) = x$ and $f(b) = y$. We have

$$xy = f(a)f(b) = f(ab),$$ so $xy \in f_*(M)$

and hence $f_*(M)$ is a submagma.

If f is a morphism of monoids, then

$$e_N = f(e_M),$$ so $e_N \in f_*(M)$ and

therefore $f_*(M)$ is a submonoid.

If f is a morphism of groups and $x \in f_*(M)$, there exists $a \in M$ such that $f(a) = x$.

We have $x^{-1} = f(a^{-1})$, so $x^{-1} \in f_*(M)$

and hence $f_*(M)$ is a subgroup. □

Notation: $f_*(M) = \mathrm{Im}(f)$.

Proposition Let $f : M \to N$ be a morphism of magmas (resp. monoids, resp. groups). For every submagma (resp. submonoid, resp. subgroup) $K \subset N$, the inverse image $f^*(K) \subset M$ is a submagma (resp. submonoid, resp. subgroup).

Proof Let $x, y \in f^*(K)$ and put

$$a = f(x), \quad b = f(y),$$

so that $a, b \in K$. As K is a submagma, we have $ab \in K$. But

$$f(xy) = f(x)f(y) = ab,$$

so $xy \in f^*(K)$ and hence $f^*(K)$ is a submagma.

Suppose f is a morphism of monoids and K is a submonoid. We have $e_M \in f^*(K)$ because $f(e_M) = e_N$ and $e_N \in K$.

Suppose f is a morphism of groups and K is a subgroup. Let $x \in f^*(K)$ and put $a = f(x)$, so that $a \in K$. As K is a subgroup, we have $a^{-1} \in K$. But

$$f(x^{-1}) = a^{-1},$$

so $x^{-1} \in f^*(K)$, and hence $f^*(K)$ is a subgroup of M. $\square$

Def Let $f : M \to N$ be a morphism of monoids (resp. groups). The submonoid (resp. subgroup) $f^*(\{e_N\})$ is called the kernel of f and denoted $\mathrm{Ker}(f)$. We have $\mathrm{Ker}(f) = \langle e_M \rangle_R$, the R-eq. class of e_M, where R is the congruence defined by f.

Proposition Let M be a magma, $T \subset M$ a subset such that $\langle T \rangle = M$.

For every morphism of magmas $f : M \to N$, we have $\langle f(T) \rangle = f_*(M)$.

(if M is generated by T, then $f_*(M)$ is generated by $f_*(T)$).

Proof Since $f_*(T) \subset f_*(M)$ and $f_*(M)$ is a submagma, we have $\langle f_*(T) \rangle \subset f_*(M)$.

If this inclusion were strict, then the inclusion

$$f^*(\langle f_*(T) \rangle) \subset f^*(f_*(M))$$

would be strict. (*) If so, the observations

- $T \subset f^*(\langle f_*(T) \rangle)$
- $f^*(\langle f_*(T) \rangle)$ is a submagma of M
- $f^*(f_*(M)) = M$

contradict the hypothesis $\langle T \rangle = M$. □

(*) For any surjection $\varphi : A \to B$ and a strict inclusion $C \subset B$, the inclusion $\varphi^*(C) \subset \varphi^*(B)$ is strict.

Proposition Let $f, g : M \to N$ be morphisms of magmas and $S \subset M$ a subset such that $\langle S \rangle = M$. If $f|_S = g|_S$, then $f = g$.

Proof (Sketch) The hypothesis $\langle S \rangle = M$ means that every element of M is of one of the forms (at least)

s $\quad (s \in S)$

or st $\quad (s, t \in S)$

or $(st)u$, $s(tu)$ $\quad (s, t, u \in S)$

or

$((st)u)v$, $(s(tu))v$, $(st)(uv)$, $s((tu)v)$, $s(t(uv))$

$(s, t, u, v \in S)$.

and so on.

The hypothesis $f|_S = g|_S$ means that f and g agree on all these expressions.

For example,

$$\begin{aligned} f(((st)u)v) &= f((st)u)\,f(v) \\ &= (f(st)f(u))\,f(v) \\ &= ((f(s)f(t))f(u))f(v) \\ &= ((g(s)g(t))g(u))\,g(v) \\ &= g(((st)u)v). \end{aligned}$$

It follows that $f = g$. □

Def Let M and N be magmas. On the product set $M \times N$, define a law of composition by

$$(x_1, y_1)(x_2, y_2) = (x_1 x_2, y_1 y_2).$$

The magma $M \times N$ is called the product of the magmas M, N. It comes equipped with morphisms of magmas

$$p_1 : M \times N \longrightarrow M,$$
$$p_2 : M \times N \longrightarrow N.$$

Prop (Universal property of the product)
Let P be a magma equipped with morphisms of magmas

$$f_1 : P \longrightarrow M,$$

$$f_2 : P \longrightarrow N.$$

There is a unique morphism of magmas

$$\varphi : P \longrightarrow M \times N$$

such that $f_1 = p_1 \circ \varphi$,

and $f_2 = p_2 \circ \varphi$.

Proof As $M \times N$ is the product of the sets M, N, there is a unique mapping

$$\varphi : P \longrightarrow M \times N$$

such that $f_1 = \phi_1 \circ \varphi$, $f_2 = \phi_2 \circ \varphi$.
It is sufficient to prove that φ is a morphism of magmas. Indeed, for any $x, y \in P$, we have

$$\begin{aligned}
\varphi(xy) &= (f_1(xy), f_2(xy)) \\
&= (f_1(x) f_1(y), f_2(x) f_2(y)) \\
&= (f_1(x), f_2(x))(f_1(y), f_2(y)) \\
&= \varphi(x)\varphi(y)
\end{aligned}$$

□

Proposition Let M, N, P be magmas. The mapping

$$\begin{aligned} \mathrm{Hom}(P, M \times N) &\longrightarrow \mathrm{Hom}(P, M) \times \mathrm{Hom}(P, N) \\ \varphi &\longmapsto (p_1 \circ \varphi, p_2 \circ \varphi) \end{aligned}$$

is a bijection of sets.

Proof This is a reformulation of the previous proposition. (the existence of φ implies surjectivity, its uniqueness implies injectivity.) $\square$

Proposition (Uniqueness of the product)

Let M, N be magmas.

If P is a magma and

$q_1 : P \to M$, $q_2 : P \to N$ are morphisms of magmas such that the triple (P, q_1, q_2) has the same universal property as the triple $(M \times N, p_1, p_2)$, then the unique morphism

$$P \xrightarrow{\varphi} M \times N \text{ such that } \begin{cases} q_1 = p_1 \circ \varphi \\ q_2 = p_2 \circ \varphi \end{cases}$$

is an isomorphism.

Proof By the universal property of the triple (P, q_1, q_2), there is a unique morphism of magmas $\varphi : M \times N \to P$ such that

$$\begin{cases} p_1 = q_1 \circ \varphi \\ p_2 = q_2 \circ \varphi \end{cases}$$

Now, $\varphi \circ \varphi : P \to P$ is a morphism of magmas satisfying

$$q_1 \circ (\varphi \circ \varphi) = (q_1 \circ \varphi) \circ \varphi = p_1 \circ \varphi = q_1,$$

and $q_2 \circ (\varphi \circ \varphi) = (q_2 \circ \varphi) \circ \varphi = p_2 \circ \varphi = q_2.$

As the morphism $Id_P : P \to P$ has the same properties, we have $\varphi \circ \varphi = Id_P$, by the universal property of (P, q_1, q_2).

Similarly, $\varphi \circ \varphi = Id_{M \times N}$ and therefore φ is an isomorphism. □

We say that the product is unique up to a unique isomorphism.

The construction of the product can be generalised to any family $(M_\lambda)_{\lambda \in \Lambda}$ of magmas or monoids or groups or modules.

Prop. If M and N are monoids, or groups, or modules, then so is the product magma $M \times N$. Moreover, for every pair $f_1 : P \to M$, $f_2 : P \to N$ of morphisms of monoids, or groups, or modules, the unique morphism of magmas $\varphi : P \to M \times N$ such that

$$f_1 = p_1 \circ \varphi \,, \quad f_2 = p_2 \circ f$$

is a morphism of monoids, or groups or modules.

Proof Indeed, if e_M, e_N are neutral elements of M, N, then (e_M, e_N) is a neutral element for $M \times N$. If f_1 and f_2 are unital morphisms then $\varphi(e_P) = (f_1(e_P), f_2(e_P)) = (e_M, e_N)$.

If $x \in M$ and $y \in N$ are invertible, then $(x, y) \in M \times N$ is invertible and $(x, y)^{-1} = (x^{-1}, y^{-1})$.

The rest of the proposition is obvious. □